

2B. PROJECT DESCRIPTION: Vision, Goals, and Driving Requirements

Vision: The Association of American Indian Physicians (AAIP) IHEART taskforce, in partnership with SGDI, envisions a national integrated data and services platform to advance data intensive and Artificial Intelligence (AI) enabled research, education and workforce development across the US. This novel initiative, IHEART National Pathways Data and AI Infrastructure (NIHDPI) fills a critical cyberinfrastructure gap for historically undeserved and geographically distributed communities, broadening participation, accelerating discovery, and strengthening a globally competitive STEM and health workforce. **Goal:** To create a novel national-scale, Indigenous-led cyberinfrastructure (CI) designed to support data-intensive and AI enabled research on the health workforce and STEM education pathways for American Indian and Alaska Native (AI/AN) learners. The long-term goal will optimize novel discoveries in various scientific research domains across national campus communities with AI prominence in computational literacy.

AAIP is the oldest and most prominent national AI/AN physician organization in the United States. Established in 1971, it serves 574 federally recognized Tribal Nations and works to motivate AI/AN students to pursue careers as physicians, scientists, and allied health professionals. AAIP's work spans decades of direct engagement with Indigenous learners, health systems, and Tribal communities, giving the organization deep knowledge of the barriers and opportunities that shape the health professions pipeline. The organization's long history and broad reach inspiring Native students to pursue medical and health-professional degrees uniquely position it to lead a national CI effort grounded in established pathways programs and aligned with community priorities.

IHEART, launched in 2021 as AAIP's dedicated pathways initiative, currently convenes a national collaborative network of more than 350 Indigenous physicians, educators, health profession students, Tribal organizations, and regional partners. This amazing network operates through five regional hubs, each contributing to local programs and high impact data collection systems shaped by regional context, culture, and community need. IHEART functions as a national pathways ecosystem; however, the data environment and cyberinfrastructure (CI) that could improve and unite students into the pathways ecosystem does not yet exist. AIAN healthcare providers clinically improve access to care and health outcomes. (St. Germaine R. & St. Germaine M., 2022)

Figure 1. IHEART Regions



Source: Indigenous Health, Education, and Resources Taskforce. (n.d.). Regions [Regional map]. IHEART Native Workforce. Retrieved December 2, 2025, from <https://www.iheartnativeworkforce.org/regions/>

SGDI, the technical collaborator responsible for designing, deploying, and operating the CI, brings professional expertise in data engineering, secure high-performance computing environments, and AI workflow development. The proposed CI initiative will operate from a dedicated SGDI facility in Minnesota, nearest to the IHEART headquarters, student communities, and medical school, providing secure, high-availability compute and storage services suited to national-scale multi-modal pathways data and GPU-enabled AI workloads.

RATIONALE: This proposal addresses the performance gap by building a production-grade, standards-driven, federated data and AI environment aligned with NSF’s Integrated Data and AI Infrastructure for Science and Engineering (IDSS) Category I expectations.

This proposed initiative strongly aligns with IHEART’s 2025 data strategic goals, specifically, the following:

(1) *Development of regional networks that promote local systems-level change focused on AI/AN health workforce that includes fostering the development of systems-focused leaders.* Infrastructure will support end-to-end pathways research, from data ingestion and harmonization through reproducible analytics, AI modeling, and ethically appropriate dissemination. **(2) *Identification and dissemination of effective systems-based solutions.*** It will embed Indigenous data sovereignty through the CARE Principles (Collective Benefit, Authority to Control, Responsibility, and Ethics) (Carroll et al., 2020; Indigenous Navigator, 2024). This is essential given the documented legacy of Indigenous data misuse, misclassification, and extraction in national research systems (Native Nations Institute, n.d.; Department of Health and Human Services, 2024; Roberts & Montoya, 2023). **(3) *The creation of a centralized national resource for health professions workforce development that centers AI/AN expertise and cultural ways.*** The IHEART infrastructure will unify and elevate Indigenous pathways data into a secure, interoperable, AI-enabled cyberinfrastructure that can support rigorous science, improve program effectiveness, and strengthen the Indigenous health workforce for decades to come (Annual 2025 AI/AN Healthcare Workforce Development Summit, 2025).

Given the alignment with AAIP and IHEART’s goals and the expedient need for AI infrastructure, the proposed project is supported by the AAIP Board President (See Letter of Collaboration #1) and IHEART Board President (See Letter of Collaboration #2) and will be executed in collaboration with SGDI, a Native-led data organization (see Letter of Collaboration #3).

Our life cycle model (St. Germaine, R. and St. Germaine, M., 2020) will bring the first data ecosystem (St. Germaine, 2022) to the Midwest geographic area for healthcare and medical professional research, retention, and recruitment. Furthermore, in reference to a unique CI environment, St. Germaine (2021) codified algorithms for analyzing small population data, eliminating the misclassification, misidentification, and outlier data designations that affect Native population data. SGDI’s researchers are uniquely experienced in Native Nation’s data analysis, workflow computation, and associated student learner environments (reservation, urban and rural environments, cultural components, and Native life).

(2) PROJECT DEFINITION AND SPECIFICATION

Project Definition: Infrastructure Design and Architecture

NIHDPI establishes a national, production-grade cyberinfrastructure that integrates multi-source Indigenous pathways data to enable reproducible AI and statistical research and to advance rigorous program evaluation and policy analysis across the United States. Designed to meet NSF IDSS Category I requirements for durable, interoperable data services tightly coupled to advanced analytic workflows, NIHDPI further contributes a technically enforced, CARE-aligned governance architecture that embeds sovereign data constraints directly into metadata, access controls, and computational execution.

At its core, the infrastructure provides an interoperable federated data fabric spanning the full pathways continuum, including K–12 preparatory programs, bridge initiatives, Tribal and non-Tribal higher education, graduate and professional health programs, residency training, and early-career workforce outcomes. National survey data indicate that approximately 6.8% (6,100) of American Indian/Alaska Native undergraduates are formally majoring in “health sciences” fields, which provides a concrete lower-bound estimate of students already clearly on health-profession pathways (King, K. et al, 2024). However, many healthcare related professional programs and colleges such as medicine, nursing, pharmacy, psychology, public health and other associated healthcare professions admit students from a wide range of majors, so long as they complete prerequisite coursework and demonstrate sustained interest in health careers. For that reason, this project uses a broader, working assumption that an estimated 10-20% (9,000 to 22,000) of AI/AN undergraduates may be in health profession relevant majors or trajectories such as biology, biochemistry, psychology, and human services are included, alongside students in designated health programs. The 6.8% figure anchors the analysis in observed federal data, while the 10–20% range captures a potentially larger pool of Native students who could realistically be supported with targeted extracurricular and academic interventions to progress into graduate or professional healthcare programs. This integrated longitudinal scope is essential for scientific inquiry into transition dynamics, retention mechanisms, intervention effects, and return-to-community trajectories.

SGDI will integrate AI utilization patterns in education and workforce science that increasingly depend on standardized data fabrics, transparent metadata, performant compute, and shared analytical pipelines to leverage predictive modeling, fairness auditing, and multi-modal inference.

The infrastructure consists of tightly coupled components that together support an end-to-end pathways research lifecycle. Table 1 summarizes the principal components.

Table 1. Core System Components

Component	Function in the CI	Scientific value
Federated data fabric	Connects multi-institution pathways datasets through standardized ingestion and harmonization	Enables longitudinal modeling across institutions and regions
On-premises compute and storage	SGDI-operated secure cluster for ETL, analytics, and AI workloads	Ensures stable, reproducible performance for national research cohorts
Standards-based APIs	Provide controlled ingestion, access, and dissemination interfaces	Enables interoperability and reuse by external research teams
Reproducible analytic and AI pipelines	Containerized workflows for prediction, NLP, geospatial inference, fairness auditing	Supports validation, comparability, and cross-disciplinary methods research
User dashboards and evaluation tools	Human-centered interfaces for program directors, educators, and researchers	Translates CI outputs into actionable pathways insights

CARE-aligned governance enforcement	Metadata tagging, role-based access, region-based sovereign tiers	Produces ethically valid and community-authorized science
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System Components: Architecture Data Life Cycle as Layers, Types of Categories, and Utilization:

The CI is organized into five functional layers that correspond to NSF’s expectations for production cyberinfrastructure. Each layer is designed to be modular, extensible, and compatible with national CI ecosystems.

Table 2. Architecture Layers and Functions

Layer	Primary functions	Implementation emphasis
1. Data ingestion and integration	Secure multi-format ingestion, automated validation, harmonization, versioned ETL	High-throughput pipelines, FAIR metadata capture
2. Storage and metadata services	Encrypted object storage and databases, distributed catalog, lineage tracking	Interoperability with CEDS, PESC, and CARE/OCAP metadata
3. Compute and AI environment	GPU-enabled ML cluster, high-core CPU batch nodes, JupyterHub, model registry	Reproducible container stacks and scheduled longitudinal workloads
4. Access and interoperability APIs	Public and governed APIs, SDKs (Python/R), federated identity via OIDC	Controlled tiers for sensitive and non-sensitive data
5. User interface and tooling	Dashboards, program evaluation modules, report generation, export services	Low-latency national access and usability for non-technical users

Leverage of the Data Model and Interoperability - Connectivity

The infrastructure will harmonize pathways datasets using open, widely adopted education and workforce standards to ensure cross-institution comparability and long-term reuse. SGDI will structure the data model to support both analytic rigor and community relevance. Key features include: (a) standardized longitudinal identifiers and stage definitions, (b) shared metric libraries for recruitment, retention, matriculation, graduation, and workforce placement, and (c) governance metadata that encodes sovereign access constraints and culturally responsive interpretation guidance.

Interoperability is achieved through standardized formats (CSV, JSON, Parquet), published schemas, and open APIs that enable integration with NSF CI resources such as ACCESS and NAIRR-like AI workflows. The system is intentionally built to serve as a domain-specific data layer that can plug into national compute ecosystems, thereby increasing the scientific return of existing NSF investments while filling a recognized gap in Indigenous pathways data infrastructure.

AI and Analytical Capabilities to Enhance Model Development (Complementarity): The CI enables a research portfolio centered on longitudinal pathways inference and fairness-aware AI. The analytical environment supports iterative model development and reproducibility through containerized execution and a model registry.

Synergistic core methodological classes include:

1. Predictive longitudinal modeling to estimate transition probabilities and risk factors across pathway stages developed by SGDI programmers.
2. Intervention effectiveness analysis using cross-regional harmonized cohorts.
3. Natural-language processing of program evidence to systematize qualitative intervention features and align them with outcomes.
4. Geospatial and socio-ecological modeling to quantify return-to-community dynamics and regional workforce effects.
5. Fairness and bias auditing to evaluate model performance across demographic and regional strata and to support AI equity research. SGDI will introduce novel Native approaches to evaluation.

Generalizability and Reuse: While the CI is centered on Indigenous health and STEM pathways, the architecture is intentionally generalizable. The federated data fabric, governance-embedded metadata design, and reusable AI workflows provide a transferable blueprint for other national workforce and education domains, especially those involving sovereign or underrepresented populations. In this way, the infrastructure contributes a novel CI pattern to the NSF ecosystem by demonstrating how national-scale data services and Indigenous data sovereignty can be operationalized together in a production environment.

(3) OPERATIONS PLAN

Overview of Operations: The IHEART NIHDPI will operate as a national-scale, production cyberinfrastructure jointly governed by AAIP and IHEART; and technically managed by SGDI. Operations begin immediately upon award. The project will achieve Full Operational Readiness by the end of Year 1, consistent with IDSS Category I requirements, and will sustain reliable national services through Year 5. The operational model is designed to support intermittent but robust use by a distributed user community of 6,100 to 22,000 registered participants, with peak research and training activity occurring in academic cycles.

Operational priorities are to (1) maintain a trustworthy and secure data environment, (2) provide stable compute and AI services for reproducible research, (3) deliver responsive national onboarding and support, and (4) implement CARE-aligned governance through continuous policy enforcement and transparent oversight.

Modes of Operation and Core Functions: Operational functions are organized into four coordinated areas. Table 1 summarizes primary activities and reliability targets.

Table 1. Core Operational Areas

Operational area	Primary activities	Reliability expectation
Data operations	Multi-region ingestion scheduling, validation, harmonization, metadata updates, lineage and version control	Continuous pipeline availability with rapid failure recovery
Compute and AI operations	GPU job scheduling, batch longitudinal analyses, interactive notebooks, model registry management, reproducibility checks	Peak-ready shared compute; stable environments for episodic research use

User access operations	Identity provisioning, tiered access enforcement, API key management, usage monitoring, export compliance checks	Secure, low-latency national access for intermittent users
Governance operations	CARE Oversight Council reviews, access request adjudication, stewardship logs, ethics compliance auditing	Documented, community-authorized decision making

Data Operations: Data operations are designed for sustained multi-source ingestion with scientific validity and governance integrity. SGDI will implement automated ingestion pipelines for structured and unstructured inputs including program datasets, institutional exports, surveys, and program documentation. Each ingestion workflow includes schema validation, de-duplication, harmonization into national standards, and governance tagging to enforce CARE-aligned access rules.

Operational tasks include scheduled ingestion (daily, weekly, or monthly depending on partner capacity), metadata refresh, schema drift monitoring, and lineage maintenance. All transformations and releases are versioned to enable reproducible longitudinal research.

Compute and AI Operations: Compute services are operated to support episodic but intensive academic research. The GPU and CPU environment will be managed through a shared queue and reservation model that prioritizes (1) national research cohorts, (2) evaluation and reporting cycles, and (3) training institutes and workshops. Containerized environments will be maintained as curated baselines to ensure reproducibility across users and years.

SGDI will operate JupyterHub for interactive analysis, schedule longitudinal batch jobs for national reports, maintain a model registry, and conduct periodic validation of model runtime environments. This approach ensures scientific repeatability without requiring continuous high utilization.

User Engagement, Onboarding, Training, and Support:

National onboarding and support are required under IDSS Category I and are central to equitable CI uptake. The onboarding pipeline is regionally coordinated by IHEART and technically provisioned by SGDI. Table 2 outlines onboarding stages.

Table 2. National Onboarding Pipeline

Stage	Lead responsibility	Description
Intake and registration	IHEART regional hubs	Program and institutional intake, initial use-case alignment
Identity provisioning	SGDI	Account creation, federated identity setup, role assignment
Governance review	CARE Oversight Council and regional stewards	Approval of access tier, sovereign constraints, data use agreements
Training assignment	IHEART and SGDI	Placement into appropriate onboarding and skills modules

Activation and support	SGDI helpdesk with regional liaisons	Ongoing technical and program evaluation support
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Training and capacity building will be delivered through monthly virtual workshops, quarterly developer bootcamps, annual in-person technical institutes, and a dedicated TCU-focused AI and data literacy series. Support is provided through a national ticketing helpdesk, scheduled office hours, regional liaisons, and documentation portals that include reproducible workflow tutorials.

Personnel for Operations: Operations require a combined technical and governance workforce. SGDI provides the technical CI team; and AAIP/IHEART provides domain governance and regional coordination. Table 3 summarizes roles.

Table 3. Operations Personnel

<i>Team</i>	<i>Key roles</i>	<i>Primary operational responsibilities</i>
<i>SGDI technical operations</i>	PI/Director, Systems Engineer, Data Engineer, AI/ML Engineer, Analyst, Security and Compliance Officer, Support Specialist, Program Manager	CI uptime, hardware and software operations, pipelines, AI services, monitoring, onboarding, helpdesk
<i>AAIP/IHEART governance</i>	National pathways director, five regional coordinators, CARE Oversight Council, advisory committee	Sovereign governance, access adjudication, community oversight, regional data coordination
<i>External advisors</i>	CI SMEs, Indigenous governance SMEs, AI ethics and workforce science experts; AI legal counsel	Annual review, readiness feedback, NSF reporting input; AI governance, AI models, cyber protections

Access and Allocation Principles

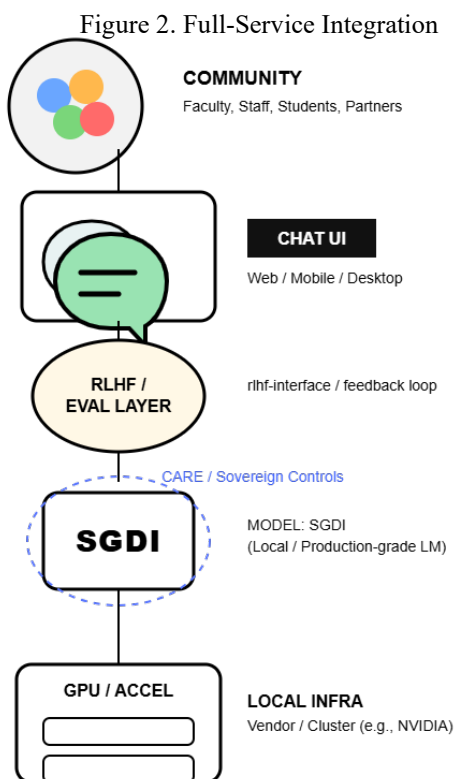
Access to CI resources follows NSF expectations for broad participation while honoring Indigenous sovereignty. The system uses tiered allocation based on data sensitivity, research merit, and CARE authorization. Table 4 defines tiers.

Table 4. Access Tiers

TIER	DATA TYPE	ACCESS CONTROL
TIER 1 PUBLIC	Non-identifiable aggregates, open dashboards, ethically permissible de-identified outputs	Open access with rate limits
TIER 2 RESEARCH RESTRICTED	De-identified structured datasets, program-level metadata	Requires research approval and governance documentation
TIER 3 SOVEREIGN GOVERNED	Sensitive pathways and demographic data	Requires CARE Council approval and regional authorization

Allocation decisions consider scientific merit, educational impact, community benefit, ethical compliance, and system load.

Figure 2 displays a flowchart of the working component outlay.



Early Operations and Transition to Full Service: Year 1 is structured to meet NSF readiness expectations. The first six months focus on hardware installation, secure environment configuration, ingesting initial partners, and implementing CARE governance tooling. Months 7 to 12 focus on cross-regional integration, load testing, security hardening, release of baseline dashboards and AI workflows, and completion of the Operational Readiness Review.

After readiness certification, Years 2 to 5 operate in full-service mode. This includes maintaining uptime targets, expanding data pipelines and user onboarding, releasing open-source CI modules, publishing national reports and open aggregates, and executing the planned hardware refresh window in Years 3 to 4.

Technology Refresh and Continuous Improvement:

Consistent with NSF Category I expectations, the CI will not remain static. SGDI will implement a rolling maintenance and refresh strategy including scheduled patching, security upgrades, and mid-award compute and storage refreshes as usage grows. AI workflows will be iteratively updated as new

longitudinal cohorts and fairness methods emerge, with reproducibility maintained through versioned containers and registry controls.

(4) PERFORMANCE OBJECTIVES AND MEASURES

Overview: Performance assessment is organized around five domains that NSF emphasizes for Category I IDSS projects: technical performance, operational reliability, user community growth, data quality and governance effectiveness, and scientific and educational impact. Metrics are selected to be measurable, auditable, and appropriate for a national cyberinfrastructure that serves a large, registered community (6,100 to 22,000 users) with intermittent but high-value research and education use. Performance will be reviewed continuously through automated telemetry and governance logs and reported annually to NSF and the broader IHEART community. Support logs and governance dashboards will be included in annual NSF reporting.

To meet the goals of advancing AI infrastructure and efficacy in research and education, NIHDPI Objectives will:

- 1) establish a robust infrastructure that advances AI, utilizing machine learning, data science, and generative adversarial networks.
- 2) advance AI through identification and dissemination of effective systems-based solutions; and
- 3) establish a centralized national resource for health professions workforce development.

OBJECTIVE 1: ESTABLISH A ROBUST INFRASTRUCTURE

Technical Performance Objectives: The infrastructure must meet production cyberinfrastructure expectations for availability, responsiveness, and throughput under national-scale use. Table 5 defines core technical targets. Qualitative and quantitative benchmarks will be reviewed by the Evaluator and PI.

Table 5. Technical Performance Objectives

<i>Performance area</i>	<i>Metric</i>	<i>Target</i>	<i>Measure and reporting</i>
<i>System availability</i>	Annual CI uptime	$\geq 99.0\%$	Automated monitoring with monthly and annual rollups
<i>Data pipeline reliability</i>	Successful ingestion jobs	$\geq 97\%$ per quarter	ETL run logs and failure audits
<i>API performance</i>	API availability	$\geq 99.0\%$	Endpoint health checks, latency dashboards
<i>Interactive analytics</i>	Median dashboard query time	< 3 seconds	Real-time usage telemetry
<i>Notebook/compute access</i>	Interactive notebook launch time	< 15 seconds	JupyterHub monitors under nominal load
<i>GPU job execution</i>	Median queued job start time	< 2 minutes under nominal load	Scheduler logs and queue statistics
<i>Storage integrity</i>	Data loss incidents	Zero	Backup validation and redundancy audits

2. Operational Reliability and Support Objectives

Operational performance ensures that users can access data and tools when needed, that issues are resolved rapidly, and that governance decisions remain transparent. Table 6 defines operational targets.

Table 6. Operational Reliability Measures

Operational function	Metric	Target
Incident response	Response time for critical outages	< 30 minutes
Issue resolution	Mean time to resolution (all tickets)	< 48 hours
Scheduled maintenance	Downtime per month	≤ 8 hours
Documentation coverage	Core services documented	100%
Governance turnaround	Governed access request decision time	< 10 business days
Governance traceability	Decisions logged and published internally	100%

Justification for Cyberinfrastructure and Equipment Investments

The requested cyberinfrastructure investments are driven by the scientific scale of the problem and the projected national user community. Indigenous pathways research requires a longitudinal, multi-institution data fabric and GPU-enabled AI environment that can support national-scale inference, reproducibility, and fairness-centered modeling. A lightly provisioned system would not meet the requirements of Category I IDSS production readiness, nor would it support a user community in the 6,100 to 22,000 range.

From a national workforce perspective, the need for rigorous pathways science is well documented. AI/AN people represent roughly 2.9 percent of the U.S. population, yet account for only about 0.3 percent of active physicians, approximately 2,500 physicians nationally (Association of American Medical Colleges, 2018; Brodt et al., 2021). Indian Health Service facilities report persistent clinical vacancy rates averaging around 30 percent, including recent estimates of 36 percent physician vacancies, with some regions reporting rates even higher (U.S. Government Accountability Office, 2022; National Indian Health Board, 2025; Indian Health Service, n.d.). These disparities are not solvable without understanding where learners enter the pipeline, where they exit, which interventions are effective, and how return-to-community patterns unfold over time. That understanding requires integrated national-scale data and the compute to analyze it.

The infrastructure must also serve a broad and distributed educational ecosystem. There are 35 accredited Tribal Colleges and Universities operating on more than 90 campuses across 15 states, and they are core contributors and users of this CI (Centers for Disease Control and Prevention, 2024; Native Nations Institute, n.d.). When combined with IHEART’s K–12 partners, Indigenous-serving colleges, professional schools, and regional hubs, the projected active national user base of 6,100 to 22,000 is realistic for a five-year production CI. System design therefore assumes sustained multi-region ingestion, concurrent research workloads, and high-availability services.

In addition, the scientific agenda depends on GPU acceleration and high-throughput storage. Predictive modeling of progression and retention, NLP of program evidence, geospatial return-to-community modeling, and fairness audits require iterative training and subgroup evaluation on harmonized multi-institution cohorts. Those operations are computationally intensive and must be reproducible, which requires a dedicated, containerized GPU and CPU environment with stable performance. Cloud bursting is not sufficient as a primary strategy because much of the data is sensitive and governed, and because CARE enforcement and stable national operations depend on controlled, sovereign-aligned infrastructure.

Table 7 links the project’s scientific drivers to concrete infrastructure requirements sized for a 6,100 to 22,000 user community.

Table 7. Scientific Drivers and Infrastructure Requirements

Scientific driver	Infrastructure requirement	Why scale matters
Longitudinal national pathways modeling	Federated data fabric, multi-petabyte storage, high-throughput ETL	National cohorts spanning K–12 through practice require durable storage and versioned integration.
AI prediction, progression, and intervention modeling, and non-parametric statistics	GPU cluster, CPU batch nodes, model registry	Multiple concurrent modeling teams require GPU availability and reproducibility tooling.

NLP and unstructured program evidence analysis	Mixed GPU and CPU NLP pipeline capacity	Thousands of documents and reports across regions must be processed consistently.
Geospatial return-to-community modeling	High-memory compute, fast object storage	Linking education and workforce data to regional context requires memory-intensive analytics.
Fairness and bias auditing	GPU acceleration, repeated subgroup evaluation pipelines	Valid fairness requires repeated model-runs using statistical multiple baseline theory across demographic and regional strata.
National dashboards and CARE program evaluation in data governance	Dedicated API servers, caching, monitoring	Thousands of simultaneous users need low-latency access to aggregates and tools.

Table 8 provides a more specific equipment and cost justification. These ranges reflect a five-year national production CI sized for 6,100 to 22,000 users, designed to meet NSF Category I readiness and up-time expectations. Budgeted expenses, estimated at \$2M, are presented as ranges because final configuration depends on vendor quotes at award.

Equipment Plan and Cost Ranges: The following equipment profile supports a national user community with intermittent access needs and limited simultaneous GPU demand. Costs are shown as ranges to reflect final vendor quotes at award, but totals are constrained to approximately \$2.0M.

Table 8. Equipment and Cost Range Justification

Equipment category	Planned capacity (moderate concurrency)	Estimated cost range	Justification tied to userbase
GPU compute servers	3 GPU servers, each with 4 data-center GPUs (12 GPUs total), for shared training and inference queues	\$700K–\$900K	Supports peak ML/NLP/fairness modeling by research cohorts without assuming constant use.
CPU compute and ETL nodes	4–6 high-core servers for ingestion, harmonization, batch analytics, and non-GPU workloads	\$300K–\$450K	Ensures continuous multi-region data pipelines and periodic large analyses.
Primary storage (object + database)	~0.8–1.2 PB usable encrypted storage with redundancy	\$350K–\$500K	Adequate for multi-year longitudinal data, documents, metadata, and derived datasets.
Metadata, database, and API layer servers	3–5 servers dedicated to catalogs, governed access, dashboards, and reporting engines	\$150K–\$250K	Maintains low-latency national access for thousands of intermittent users.
Networking and security hardware	Secure transfer gateways, firewalls, IAM integration, 25–100G internal backbone	\$100K–\$180K	Required for federated ingestion and CARE-aligned access tiers.
Monitoring, backup, redundancy	Telemetry stack, immutable backups, failover storage targets	\$120K–\$200K	Supports production readiness and national reliability expectations.

Estimated equipment total: \$1.72M–\$2.48M

Planned target at award: ~\$2.0M, with final configuration selected to stay within cap.

This equipment will meet how SGDI will develop the CI to further AI research:

- The national dashboard userbase could potentially reach 22,000 users. Most users will interact through dashboards, secure data queries, or limited exploratory analysis (EA). These activities depend primarily on a stable data, metadata, and API core.
- A smaller active research subset performs GPU training or advanced modeling during defined windows (semester projects, summer institutes, grant evaluation cycles). Three modern GPU servers with 12 total GPUs provide robust shared access for these peaks.
- Longitudinal pathways science is storage-dominant and governance-dominant more than continuously compute-dominant. The proposed 0.8–1.2 PB storage tier ensures durability, lineage, and release capacity without over-buying compute.

This right-sized approach allows the CI to support national-scale scientific merit now, while keeping a clear growth pathway. If GPU demand expands beyond episodic peaks, SGDI can add nodes through refresh cycles or partner with external NSF CI for overflow on ethically appropriate, non-sensitive workloads.

OBJECTIVE II. ADVANCE AI THROUGH IDENTIFICATION AND DISSEMINATION OF EFFECTIVE SYSTEMS-BASED SOLUTIONS

3. User Growth and Community Engagement Objectives

User growth targets reflect a broad national registration base with increasing levels of active engagement: 6,100–22,000 range as a reasoned estimate built from (a) ~107,000 AI/AN undergraduates nationally, (b) the share of AI/AN degrees in STEM/health fields, and (c) general patterns of AI/AN onboarded in health majors. Table 3 provides annual targets for registered users and institutions onboarded.

Table 9. Comparison of National User Growth Targets and Community Engagement Measures

National User Growth Targets			Community Engagement Measures	
Year	Users	Onboarded	Engagement area — Metric — Target	
Year 1	1,000	25	Training delivery	Workshops per year — ≥ 12
Year 2	3,000	45	Advanced capacity building	Developer bootcamp participants annually — ≥ 150
Year 3	9,000	70	TCU technical series	TCU-focused AI training sessions per year — ≥ 6
Year 4	15,000	95	Onboarding completion	New users completing onboarding modules — $\geq 70\%$
Year 5	22,000	120	User satisfaction	Mean satisfaction rating — $\geq 4.5/5$

These measures emphasize equitable access and sustained usability rather than continuous compute utilization.

4. Data Quality, Integration, and Governance Objectives

Scientific validity depends on high data quality and consistent governance enforcement. Metrics in Table 10 are monitored continuously through ingestion logs and catalog audits.

Table 10. Data Quality and Integration Measures

<i>Data quality area</i>	<i>Metric</i>	<i>Target</i>
<i>Schema validation</i>	Successful schema checks per ingestion	$\geq 97\%$
<i>Harmonization accuracy</i>	Harmonization error rate	$< 1\%$
<i>Duplicate control</i>	Duplicate record rate	$< 0.5\%$
<i>Metadata completeness</i>	Required metadata fields populated	$\geq 95\%$
<i>Lineage integrity</i>	Datasets with full provenance trails	100%

SGDI PI and Evaluator will assess the governance area through CARE-aligned indicators in Table 11.

Table 11. CARE Governance Performance Measures

<i>Governance area</i>	<i>Metric</i>	<i>Target</i>
<i>Documentation</i>	Percent of governance decisions recorded	100%
<i>Compliance audits</i>	Annual audit success rate	100%
<i>CARE adoption</i>	Institutions adopting CARE-aligned practice	≥ 50 by Year 5
<i>Ethics training</i>	Users completing ethics training before Tier 3 access	100%

These measures ensure that access integrity and data sovereignty are enforced as technical CI functions, not optional policies.

OBJECTIVE III. ESTABLISH A CENTRALIZED NATIONAL RESOURCE FOR HEALTH PROFESSIONS WORKFORCE DEVELOPMENT.

5. Scientific and Educational Impact Objectives: The CI is evaluated not only as infrastructure but as a catalyst for research and education innovation. Impact targets are defined conservatively to reflect intermittent use with high scientific payoff. Table 12 summarizes research productivity measures.

Table 12. Scientific Impact Measures

Impact area	Metric	Target by Year 5
Publications	Peer-reviewed papers supported by CI	≥ 40
AI methods	Open-source AI/ML models released	≥ 10
Data products	Non-sensitive aggregate datasets released	≥ 20
Collaborations	Interdisciplinary research projects launched	≥ 25

Education and pathways impact metrics are shown in Table 13.

Table 13. Educational and Pathways Research Measures

Education/pathways area	Metric	Target by Year 5
New evaluation studies, benchmark studies Social Cognitive Career Theory, Nomothetic Theory,	Cross-regional pathway studies enabled	≥ 30

Institutional tool use	Programs using dashboards for evaluation	≥ 100 institutions
Longitudinal tracking capability	Increase in tracking completeness across stages	$\geq 60\%$
Return-to-community modeling	Regions with deployable workforce models	≥ 10

Measurement Methods and Reporting

Performance will be evaluated through three coordinated methods. Automated CI telemetry using standard monitoring tools to report uptime, latency, throughput, scheduler performance, and storage integrity. Governance and access tracking through CARE Council decision logs, tiered access dashboards, and annual ethics audits. User and impact assessment via biannual surveys, focus groups with TCU and regional partners, utilization summaries, and bibliometric tracking of CI-supported publications.

Annual reporting to NSF will include tables aligned with these objectives, trend summaries, corrective actions where targets are not met, and evidence of continuous improvement.

(5) PROJECT MANAGEMENT

i. Our Team Leadership and Management Structure

The project is managed through a three-part partnership structure that aligns domain leadership, Indigenous governance, and technical cyberinfrastructure execution. AAIP serves as the lead institution and award recipient, providing national oversight, fiscal stewardship, and alignment with the organization's long-standing mission to strengthen the AI/AN health workforce. IHEART functions as the programmatic and governance lead, coordinating regional participation, pathways use cases, and CARE-aligned oversight for data access and ethical use. SGDI serves as the technical lead and collaborator responsible for the design, deployment, and operation of the national data and AI cyberinfrastructure.

Scientific and technical direction is provided by the PI and Technical Director, SGDI Researcher St. Germaine, who is accountable for ensuring that the infrastructure and AI deliverables remain aligned with IDSS Category I requirements, are methodologically rigorous, and are delivered on schedule. This management structure ensures that the cyberinfrastructure remains both scientifically credible and community-authorized at national scale.

Roles and Responsibilities: Project execution is organized around an integrated technical, governance, evaluation, and operations workforce. Table 14 summarizes key personnel, level of effort, and responsibilities.

Table 14. Key Personnel and Responsibilities

Role	Name	FTE	Primary responsibilities
Principal Investigator Technical Director	R. St. Germaine, PhD, MPH	1.0	Provides overall scientific and technical leadership; directs AI and analytics architecture; oversees methodological rigor, governance compliance, technical risk, hiring, and NSF reporting.

Lead Data Engineer	M. St. Germaine, CSO	1.0	Leads data ingestion, harmonization, metadata standards, quality control, versioning, and privacy-preserving access controls.
Data Scientist	J. GrayEagle	1.0	Develops and validates longitudinal and AI models; performs fairness assessment, interpretability analyses, and stakeholder-facing reporting.
Data Scientists (2)	R. St. Germaine, PhD, MPH, ENPH	2.0 total	Develop parallel modeling modules and robustness testing; conduct scenario analyses; maintain reproducibility and documentation.
Machine Learning Operations Engineer	T. Gibbs	1.0	Build CI/CD workflows for models; manage registries, retraining triggers, drift monitoring, and secure production deployment.
Software Engineer III	TBD	0.5	Architects core platform services and APIs; enforces scalability and security standards; mentors junior staff.
Software Engineer Intern	TBD	0.25	Supports targeted coding, QA, documentation, UI components, and integration sprints during peak development periods.
Regional Lead Coordinator	B. Butcher, MD	0.5	Provides clinical and domain oversight; coordinates regional partners; validates relevance and interpretability of analytic outputs; supports dissemination.
Evaluator	V. LaJoie, MS	1.0	Conducts independent process and outcome evaluation; maintains metrics tied to aims; produces annual evaluation reports.
Project Manager	TBD	1.0	Oversees workplan execution, scheduling, milestone and risk tracking, partner coordination, and reporting cadence.
Administrative Assistant	TBD	1.0	Provides operational support including scheduling, travel, purchasing, compliance documentation, and communications.

This staffing profile provides sufficient depth for concurrent infrastructure buildout, multi-stream AI modeling, governance enforcement, and continuous evaluation across five years.

Work Breakdown Structure

The work plan follows a structured WBS suitable for NSF reporting and readiness review. Table 15 summarizes Level 2 work domains and their relationship to project deliverables.

Table 15. Level 2 Work Breakdown Structure

WBS Domain	Scope of work
1. Project management and governance	Planning, scheduling, reporting, fiscal oversight, CARE policy design, advisory committee coordination
2. CI architecture and deployment	Hardware procurement and installation, secure network configuration, storage and compute setup, monitoring and redundancy
3. Data pipeline development	Ingestion workflows, harmonization services, metadata schema and catalog, data quality automation

4. AI and analytics development	Baseline and advanced longitudinal models, NLP modules, geospatial models, fairness auditing pipelines
5. User tools and interfaces	Dashboards, program evaluation toolkit, export/reporting engines, IAM user portals
6. Training and user support	Documentation, onboarding modules, workshops and bootcamps, TCU technical series, helpdesk services
7. Operations and sustainability	Continuous CI operations, performance tuning, technology refresh, national dissemination, sustainability planning

Broader Impacts.

This project delivers promise of significant societal benefits by establishing an Indigenous-led, national data and AI CI to frame and strengthen AI/AN education-to-workforce pathways in healthcare and STEM. Intellectual merit: By integrating multi-institutional pathways data into a governed federated data fabric and coupling these resources with reproducible AI/ML workflows, the IHEART NIHDPI will enable robust, longitudinal analyses of where AI/AN learners enter, persist, and exit the health professions pipeline. The resulting evidence base will support more effective recruitment, mentoring, bot-locator technology to improve financial support systems, grant-seeking locators from bot technology, new curricular design, and synergistic clinical placement strategies, directly addressing persistent shortages and high vacancy rates in Indigenous-serving health systems.

A core broader impact is advancing Indigenous data sovereignty and ethical AI at national scale. NIHDPI operationalizes the CARE Principles for Indigenous Data Governance (Collective Benefit, Authority to Control, Responsibility, and Ethics) through technically enforced tiered access, community-authorized metadata constraints, and transparent oversight structures. Embedding CARE in the daily operation of cyberinfrastructure, rather than treating it as a policy add-on, provides a transferable model for conducting high-performance, data-intensive science with sovereign and historically rich populations in ways that are ethically valid and community-benefiting.

NIHDPI will also expand Indigenous participation in data-driven careers through a structured training ecosystem that includes workshops, developer bootcamps, annual institutes, and TCU-focused AI and data literacy programming. TCUs play a recognized national role in broadening Native participation in STEM and in aligning technical training with Tribal priorities; this project strengthens that role by providing access to production-grade cyberinfrastructure and real-world data science practice. While NIHDPI is centered on healthcare pathways, the skills and tools it develops will support AI/AN learners entering additional professions that increasingly rely on data science and AI, including computer science and software engineering, economics and finance, business and entrepreneurship, education and policy research, and legal or governance careers involving data rights and technology regulation.

Five-Year Timeline (Tasks Down, Quarters Across)

Table 16. Timeline and Major Milestones

# / Task Area (Lead roles)	Y1 Q1	Y1 Q2	Y1 Q3	Y1 Q4	Y2 Q1	Y2 Q2	Y2 Q3	Y2 Q4	Y3 Q1	Y3 Q2	Y3 Q3	Y3 Q4	Y4 Q1	Y4 Q2	Y4 Q3	Y4 Q4	Y5 Q1	Y5 Q2	Y5 Q3	Y5 Q4
1. Hardware specs, quotes, purchase (PI/TD, SE, PM/AA)	X																			
2. Hardware install, CI base deployment (SE, LDE, Machine Learning Ops)		X																		
3. Security, network, IAM, monitoring stack (SE, Machine Learning, PI/TD)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
4. Ingestion templates + partner intake workflow (LDE, PM/AA, CARE/IH)	X	X			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
5. ETL pipelines for initial partners (LDE, DS)		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
6. Harmonization schemas + national data model (LDE, DS, CARE/IH)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
7. Metadata catalog v1 (LDE, SE)		X	X	X	X				X				X				X			
8. Baseline AI models (progression, retention) (DS, Machine Learning)			X	X	X	X			X	X			X	X			X	X		
9. Advanced AI modules (return-to-community, geo, NLP) (DS, Machine Learning Ops, RLC)				X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
10. Fairness/bias auditing pipelines (DS, Machine Learning Ops)			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
11. Dashboards/tooling prototype (SE, DS, EVAL)		X	X																	
12. Dashboards national release + iterations (SE, DS, EVAL)			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
13. CARE governance workflows + automation (CARE/IH, PI/TD, AAIP)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
14. Pilot onboarding + training launch (PM/AA, CARE/IH, SE)	X	X	X	X																

# / Task Area (Lead roles)	Y1 Q1	Y1 Q2	Y1 Q3	Y1 Q4	Y2 Q1	Y2 Q2	Y2 Q3	Y2 Q4	Y3 Q1	Y3 Q2	Y3 Q3	Y3 Q4	Y4 Q1	Y4 Q2	Y4 Q3	Y4 Q4	Y5 Q1	Y5 Q2	Y5 Q3	Y5 Q4
15. National onboarding, workshops, institutes (PM/AA, CARE/IH, DS)			X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
16. Operational readiness prep + ORR (PI/TD, PM/AA, EVAL, AAIP)			X	X	X															
17. National scale-up of institutions/users (PI/TD, CARE/IH, PM/AA)				X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
18. CI interoperability with NSF ecosystems (PI/TD, SE, LDE)								X	X	X	X	X	X	X	X	X	X	X	X	X
19. Technology refresh / capacity tuning (PI/TD, SE, LDE)									X	X			X	X			X	X		
20. Evaluation, annual reporting, impact studies (EVAL, PI/TD, RLC)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
21. Sustainability + transition planning (PI/TD, AAIP, CARE/IH)													X	X	X	X	X	X	X	X

Table 17. Responsibility Matrix for Timeline Tasks

Timeline task area	Lead role(s) responsible	Supporting roles
Hardware specification and purchasing (Y1 M0–3)	PI/Technical Director; Lead Systems/Software Engineer III	Lead Data Engineer; Project Manager; Security and Compliance Officer
Hardware installation and CI base deployment (Y1 M4–6)	Lead Systems/Software Engineer III; Lead Data Engineer	Machine Learning Engineer; Project Manager; Administrative Assistant
Ingestion pipeline buildout and harmonization (Y1 M4–12, ongoing)	Lead Data Engineer	Data Scientists; Regional Lead Coordinator; Evaluator
AI/ML model development and validation (Y1 M7 onward)	Data Scientist (J. GrayEagle); Data Scientists (R. St Germaine)	Machine Learning Engineer; Lead Data Engineer; Regional Lead Coordinator
Governance workflow implementation and CARE enforcement (Y1 M0 onward)	PI/Technical Director; IHEART CARE Oversight Council	Regional Coordinators; Security and Compliance Officer; Project Manager
Dashboard/tool development and user testing (Y1 M7 onward)	Software Engineer III; Data Scientists	Software Engineer Intern; Evaluator; Regional Lead Coordinator
National onboarding, training, and helpdesk (Y1 M7 onward)	Project Manager; Support Specialist (SGDI)	Administrative Assistant; IHEART Regional Coordinators; Evaluator
Operational readiness review and reporting (Y1 M10–12 and annually)	PI/Technical Director; Project Manager	Evaluator; Lead Data Engineer; AAIP/IHEART leadership
National scaling and sustainability (Y2–Y5)	PI/Technical Director; AAIP/IHEART leadership	Project Manager; Evaluator; Technical leads; CARE Council

RISK MANAGEMENT AND MITGATIONS

Risk management adheres to technical and governance reviews through a maintained risk register overseen by the Project Manager and PI. SGDI will monitor all tasks and processes associated with installations, upgrades, systems changes, and primary risks documented in public-facing policies and internal documents. SGDI PI and IHEART PI will monitor any contingencies through project management approach of Plan, Do, Study, Act for onboarding and staffing changes to ensure seamless workflow and subsequent mitigations are summarized in Table 18. SGDI will keep all software and systems updated with consultation with hardware options (GPU plus server environments, networking components), in compliance with Build America, Buy America Act.

Table 18. Key Risks and Mitigation Strategies

Risk area	Risk	Mitigation strategy
Technical performance	Hardware failure or compute bottlenecks during peak research cycles	Redundant storage and failover services; monitored capacity thresholds; scheduled refresh window in Years 3–4
Data variability	Heterogeneous formats and inconsistent partner reporting	Standardized templates; automated schema validation; harmonization pipelines; regional data steward support
AI validity	Model drift or fairness degradation over time	Machine Learning Operations monitor for drift and subgroup performance; retraining triggers; annual model audits
Operational adoption	Uneven user onboarding or low engagement in some regions	Region-based onboarding supported by IHEART; targeted TCU training; helpdesk and liaison support
Governance integrity	Access disputes or delays under CARE oversight	Workflow-based adjudication; documented decision timelines; transparent governance logs
Data misuse	Risk of inappropriate access or interpretation	Tiered access controls; mandatory ethics training; audit trails; enforcement by CARE Council

Quality Assurance, Security, and Trustworthiness: Quality assurance is maintained via version-controlled, automated pipelines; reproducible containerized environments; standardized code review; and continuous monitoring against defined performance objectives. Security and trustworthiness are operationalized through encryption at rest and in transit, role- and region-based access controls, multifactor authentication, and continuous logging with auditable trails. CARE-aligned governance is embedded in metadata and access enforcement to ensure sovereign constraints are applied automatically and consistently. Assurance is sustained through quarterly vulnerability reviews and annual external advisory oversight. **SGDI Data Security Controls (selected technical measures):**

- Full-disk encryption for desktops and laptops
- Encryption of data in transit
- Encryption tools for removable media
- Host-based firewalls (desktop/laptop)
- Antivirus and anti-malware protection
- Multifactor authentication (MFA)
- Automated patch management and vulnerability scanning
- Strong physical, environmental, network, and perimeter safeguards
- Intrusion detection and prevention systems (IDPS)
- Continuous monitoring and threat detection systems